DECENT COACHING CLASSES

X SB

Mind Your Math - IX

ANSWER KEY & MARKING SCHEME

Time: 2 Hours **Maximum Marks: 40**

Q.1

Q.1 (A) [1 mark each]

(1) **(b) 9 : 7** – ratio of sides = $\sqrt{(81/49)} = 9/7$

(2) **(b) (8, 15, 17)** – $8^2 + 15^2 = 64 + 225 = 289 = 17^2$

(3) **(c) 12 cm** – $PT^2 = OP^2 - r^2 = 169 - 25 = 144$

(4) **(b) 2** – slope $= (7 - 3)/(4 - 2) = 2$

Q.1 (B) [1 mark each]

(1) $\cos\theta = \sqrt{(1 - 9/25)} =$ **4/5**

(2) Opposite angles of a cyclic quadrilateral are supplementary: $\angle C = 180° - 75° =$ **105°**

(3) $d = \sqrt{[(4 + 2)^2 + (-5 - 3)^2]} = \sqrt{(36 + 64)} =$ **10**

(4) Arc $= (60/360) \times 2 \times 3.14 \times 12 =$ **12.56 cm**

Q.2

Q.2 (A) [2 marks each – any two]

(1) AD/DB = **AE/EC** [1]; $3/5 = 6/EC \Rightarrow EC =$ **10** [1]

(2) Side opposite 30° = ½ × **hypotenuse (AC)**; AB = ½ × 14 = **7 cm** [1]; BC = $(\sqrt{3}/2) \times 14 =$ **$7\sqrt{3}$ cm** [1]

(3) $x = (2 \times 7 + 1 \times 1)/3 = 15/3 =$ **5** [1]; $y = (2 \times 8 + 1 \times 2)/3 = 18/3 =$ **6**; **P(5, 6)** [1]

Q.2 (B) [2 marks each – any two]

(1) A(small)/A(large) = (side/side)$^2 \Rightarrow 25/36 = (10/x)^2$ [1] $\Rightarrow 10/x = 5/6 \Rightarrow x =$ **12 cm** [1]

(2) AP × PB = CP × PD (intersecting chords) $\Rightarrow 6 \times 4 = 3 \times PD$ [1] $\Rightarrow PD =$ **8** [1]

(3) Slope $= (-2 - 4)/(5 - (-3))$ [1] $= -6/8 =$ **$-3/4$** [1]

(4) LHS $= \cos^2\theta \times \sec^2\theta$ [1] (since $1 + \tan^2\theta = \sec^2\theta$) $= \cos^2\theta \times 1/\cos^2\theta = 1 =$ RHS [1]

Q.3

Q.3 (A) [3 marks – any one]

(1) $\angle OAP =$ **90°**, $\angle OBP =$ **90°** [1]; $\angle AOB = 360° - (90° + 90° + 70°)$ [1] = **110°** [1]

(2) $\theta =$ **90**: arc $= (90/360) \times 2 \times (22/7) \times 7 =$ **11 cm** [1½]; area $= (90/360) \times (22/7) \times$ **49** = **38.5 cm²** [1½]

Q.3 (B) [3 marks each – any three]

(1) Apollonius theorem: $AB^2 + AC^2 = 2AD^2 + 2BD^2$ [1]; $121 + 49 = 2AD^2 + 2(49)$ [1] $\Rightarrow 2AD^2 = 72 \Rightarrow AD^2 = 36 \Rightarrow$ **AD = 6** [1]

(2) Given: PA and PB are tangent segments from P to circle with centre O. To prove: PA = PB. [½] Construction: join OA, OB, OP. [½] OA ⊥ PA and OB ⊥ PB (tangent ⊥ radius) so △OAP and △OBP are right-angled. [½] OA = OB (radii), OP common (hypotenuse). [½] ∴ △OAP ≅ △OBP (hypotenuse–side test). [½] ∴ PA = PB (c.s.c.t.). [½]

(3) $AB = \sqrt{(16 + 9)} = 5$ [1]; $BC = \sqrt{(16 + 9)} = 5$ [½]; $AC = \sqrt{(64 + 36)} = 10$ [½]; AB + BC = AC [½] ⇒ points are **collinear** [½]. (Alternative: slope AB = 3/4 = slope BC with B common.)

(4) LHS $= [(1 - \sin\theta) + (1 + \sin\theta)] / [(1 + \sin\theta)(1 - \sin\theta)]$ [1] $= 2/(1 - \sin^2\theta)$ [1] $= 2/\cos^2\theta =$ **$2\sec^2\theta$** = RHS [1]

Q.4

Q.4 [4 marks each – any two]

(1) DE ∥ BC ⇒ ∠ADE = ∠ABC and ∠A common ⇒ △ADE ~ △ABC (AA) [1]; AD/AB = 3/7 [1]; A(△ADE)/A(△ABC) = $(3/7)^2 = 9/49$ [1]; A(△ADE) : A(trapezium DBCE) = 9 : (49 − 9) = **9 : 40** [1]

(2) Let the building be 24 m high. Angle of depression of the foot = 45° ⇒ horizontal distance = 24/tan 45° = **24 m** [1½]; height of tower above the building top = 24 × tan 60° = 24√3 = 24 × 1.73 = **41.52 m** [1½]; total height of tower = 24 + 41.52 = **65.52 m** [1]

(3) Volume of cone = $(1/3)\pi r^2 h = (1/3)\pi(36)(24) = 288\pi$ cm³ [1½]; volume of sphere = $(4/3)\pi R^3$ [½]; $(4/3)\pi R^3 = 288\pi \Rightarrow R^3 = 216$ [1] ⇒ **R = 6 cm** [1]

Q.5

Q.5 [4 marks – any one]

(1) Steps: (a) Draw BC = 6 cm; at B draw ∠B = 60° and mark A with BA = 5 cm; join AC → △ABC. [1] (b) At B draw an acute angle ∠CBX below BC; mark 7 equal parts $B_1 \ldots B_7$ on ray BX. [1] (c) Join B_5 to C; draw $B_7C' \parallel B_5C$ meeting BC extended at C′. [1] (d) Draw C′A′ ∥ CA meeting BA extended at A′. △A′BC′ is the required triangle (BC′ = 7/5 × 6 = 8.4 cm, BA′ = 7/5 × 5 = 7 cm). [1]

(2) Steps: (a) Draw circle of radius 3 cm with centre O; mark P with OP = 7 cm. [1] (b) Draw the perpendicular bisector of OP; let M be the midpoint of OP. [1] (c) With centre M and radius MO = MP, draw a circle cutting the given circle at A and B. [1] (d) Join PA and PB – these are the required tangents (each of length $\sqrt{(49 - 9)} = \sqrt{40} \approx 6.3$ cm). [1]

— End of Answer Key —